Minghan Wang

https://github.com/Minghan-Wang/Google-Dino-Game-Team-8-in-STARS
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Education

Purdue University Expected Graduation: Dec 2026
Bachelor of Science in Computer Engineering (West Lafayette, Indiana) *GPA: 3.98 / 4.00*

- Certifications: Semiconductor Fabrication 101 Certificate, Summer Training, Awareness, and Readiness for Semiconductors Certificate
- Relevant Coursework: Data Structures and Algorithms, Introduction to Digital System Design, Advanced C Programming, Discrete Mathematics, Electrical Engineering Fundamentals II

Experience

Summer Training on Awareness and Readiness for Semiconductors (STARS) May 2024 – July 2024
Hardware Engineer *West Lafayette, Indiana*

- Built fundamentals of basic electronics, digital logic, register transfer level design, and verification.
- Led team to write a Google Dino game on a chip, resulting in a functional hardware prototype.
- Utilized **SystemVerilog** to write key game modules (Random Number Generator, Score Counter, Collision Detector, and Parallel and Serial Image Generator).
- Integrated modules into a top-level test bench, ensuring seamless functionality on an FPGA with ili9341 display and Seven-Segment Displays.
- Presented project progress at three stages and collaborated on designing the project documentation.

Projects

Project 1 | **System on Chip Extension Technologies (SoCET) – Digital Design** *Team Member* May 2025 – Present

- Validate tapeout flow & configuration for upcoming AFTx08 chip using AFTx07 test platform, identifying critical pre-fabrication issues to reduce cost/risk.
- Optimize CPU performance by benchmarking cache configurations (1-8KB d/i-cache), memory latency models (4-132 cycles), and RISC-V ISA extensions (RV32I/E, zba/zbb/zbs).
- Develop & present benchmarking results using **Embench-IoT** to guide data-driven decisions for tapeout.

Project 2 | **System on Chip Extension Technologies (SoCET) – Design Flow** *Team Member* Jan 2025 – Present

- Produce, optimize and verify the final physical design our chip AFTx08 by this Summer.
- Perform pre-layout Design Rule Check (**DRC**) verification by uing MITLL PDK and Cadence tools to ensure layout manufacturability.
- Prepare the final GDSII layout for submission to Skywater for tape-out.
- Develop and document a robust and scalable design flow for future projects.

Project 3 | **System on Chip Extension Technologies (SoCET) – Test Engineer** *Team Member* Aug 2024 – Dec 2024

- Work with Synopsys mentors to study Design for Test and Automatic Test Pattern Generation concepts.
- Implement scan chain insertion and performing ATPG on a PCI code design for improving controllability and observability.
- Research on DFT methodologies to enable test features on RISCV SoC and to incorporate an On-Chip Clock Controller into existing designs.
- Perform pre-DFT and post-DFT DRC checks, debugging and fixing DFT DRC violations, and performing top-down and bottom-up scan insertion.

Technical Skills

Hardware: RTL Design, FPGA Prototyping, Power Analysis, Synthesis (Genus), Tapeout Flows, Benchmarking, Cache/Memory Optimization.
Standards: RISC-V ISA.
Tools: Cadence Flowtool, Cadence Vituoso, Cadence Innovus, Cadence Genus, Simens Calibre, Embench-IoT, Open FPGA Toolchains, Git, openLane, Fusion Compiler.
Languages: SystemVerilog, C, Python.